

SearchIQ – Search Typeahead System

Project Report

Course Assignment: Build a Search Typeahead System

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1. Architecture Diagram & Explanation



Component Responsibilities

Component	File	Role
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Flask Server	<code>`app.py`</code>	Routes, startup orchestration, latency tracking
Trie	<code>`trie.py`</code>	$O(\text{prefix_len})$ suggestion lookup with per-node top-10 heap
Consistent Hash Ring	<code>`consistent_hash.py`</code>	Maps prefix keys to cache nodes
Distributed Cache	<code>`cache.py`</code>	4 in-memory cache nodes with TTL + LRU eviction
Database	<code>`database.py`</code>	SQLite with WAL mode for persistent query counts
Batch Writer	<code>`batch_writer.py`</code>	Buffers writes; flushes aggregated deltas periodically
Trending Tracker	<code>`trending.py`</code>	Exponential-decay recency scoring
Dataset Generator	<code>`generate_dataset.py`</code>	Creates 120,000+ synthetic queries (Zipf distribution)

2. Dataset Source & Loading Instructions

Dataset

- Type: Synthetically generated (no licensing restrictions)
- File: data/queries.csv
- Size: 120,000 queries
- Format: query,count (CSV with header)
- Top sample rows:

query	count
python tutorial	4,820,000
javascript tutorial	3,940,000
iphone	2,650,000
machine learning tutorial	1,820,000
java tutorial	1,340,000

Distribution

Counts follow Zipf's law: $\text{count}(\text{rank}) \approx 5,000,000 / \text{rank}^{1.1}$ with $\pm 20\%$ random noise. This mirrors real-world search frequency distributions where a small number of queries dominate traffic.

Domains Covered

Tech products, programming tutorials, e-commerce, entertainment, health & wellness, general knowledge — 6 domain pools, each expanded with 22 suffix variants ("review", "price", "2024", "near me", etc.).

Loading Instructions

```
# Step 1: Generate the dataset (creates data/queries.csv)
python generate_dataset.py

# Step 2: On first server start, data auto-loads into SQLite
python app.py
# → "Loading dataset from data/queries.csv..."
# → "Dataset loaded: 120000 queries"
# → "Trie built in ~8s (120000 entries)"
```

The server checks if the DB is already populated on each startup and skips re-loading if data exists.

3. API Documentation

`GET /suggest`

Fetch prefix-matching suggestions.

Parameter	Type	Required	Description
<code>q</code>	string	Yes	The prefix to search
<code>mode</code>	string	No	<code>basic</code> (historical count) or <code>trending</code> (recency-aware). Default: <code>basic</code>

Response:

```
{
  "suggestions": [
    {"query": "python tutorial", "score": 4820000.0},
    {"query": "python for beginners", "score": 1230000.0}
  ],
  "source": "cache",
  "latency_ms": 0.31
}
```

source is one of cache, trie, or empty.

``POST /search``

Submit a search query. Returns dummy response and triggers batch write + trending update.

Request body:

```
{ "query": "python tutorial" }
```

Response:

```
{ "message": "Searched", "query": "python tutorial" }
```

``GET /cache/debug``

Inspect consistent-hashing routing for a given prefix.

Parameter	Type	Required	Description
<code>`prefix`</code>	string	Yes	Prefix to inspect

Response:

```
{
  "prefix": "py",
  "assigned_node": "cache-node-3",
  "cache_hit": true,
  "cached_results": [...],
  "ring_info": {
    "key_hash": 282429536481,
    "node_virtual_positions": 150,
    "all_nodes": ["cache-node-1", "cache-node-2", "cache-node-3", "cache-node-4"]
  },
  "node_stats": { "hits": 42, "misses": 8, "hit_rate": 0.84 },
  "all_node_stats": [...]
}
```

``GET /trending``

Returns top-10 queries ranked by recent activity.

Response:

```
{
  "trending": [
    { "query": "python tutorial", "recent_score": 4.87, "recent_count": 5 },
    { "query": "java tutorial", "recent_score": 2.91, "recent_count": 3 }
  ]
}
```

``GET /stats``

Live system metrics.

Response:

```
{
  "db":      { "write_count": 12, "read_count": 340 },
  "cache":   { "total_hits": 890, "total_misses": 110, "overall_hit_rate": 0.89 },
  "batch":   { "total_events_received": 60, "total_batches_flushed": 2,
               "total_db_writes_saved": 58, "pending_in_buffer": 3 },
  "latency": { "samples": 1000, "avg_ms": 0.8, "p50_ms": 0.4, "p95_ms": 3.1 }
}
```

`POST /batch/flush`

Manually trigger a batch flush. No body required.

Response: { "message": "Flush triggered", "stats": {...} }

4. Design Choices & Trade-offs

4.1 Trie with Per-Node Top-K Heap

Choice: Each Trie node maintains a min-heap of the top-10 (score, query) pairs passing through it.

Why: Pure Trie traversal to find all matches under a prefix is $O(\text{subtree_size})$, which is catastrophically slow for 120,000 queries. By caching the top-10 at every node during insertion, suggestion lookup becomes $O(\text{prefix_len})$ — just traverse to the prefix node and read the heap.

Trade-off: Higher memory usage (~ 10 entries $\times$ every node) and more expensive insertions ($O(\text{prefix_len} \times \log 10)$). Acceptable since inserts happen only at batch flush time, not per request.

4.2 Distributed Cache with Consistent Hashing

Choice: 4 logical cache nodes behind a Consistent Hash Ring with 150 virtual nodes each.

Why consistent hashing over modular hashing: If we used $\text{hash}(\text{prefix}) \% N$, adding or removing a node would remap $\sim 100\%$ of keys, causing a thundering-herd problem. Consistent hashing remaps only $1/N \approx 25\%$ of keys when a node is added/removed.

Virtual nodes: With only 4 nodes and no virtual nodes, the ring would be non-uniform (some nodes covering 40% of the ring, others 10%). 150 virtual nodes per node gives $\sim 25\% \pm 3\%$ load distribution.

TTL = 300s: Stale suggestions don't persist longer than 5 minutes even if cache invalidation misses a key. Short enough for freshness, long enough to give good hit rates.

Trade-off: Cache is in-memory (single process). A real distributed system would use Redis or Memcached. For this assignment, logical nodes in one process demonstrate the algorithm correctly.

4.3 Trending Search: Exponential Decay

Formula:

$$\text{score}(q) = 1.0 \times \text{all_time_count}(q) + 5000 \times \sum e^{(-\lambda \times \text{age}_i)}$$

where $\lambda = \ln(2) / 600$ (half-life = 10 minutes).

Why exponential decay over a simple time window: A sliding window treats all events within the window equally — a query searched 59 minutes ago counts the same as one searched 1 minute ago. Exponential decay naturally weights recency without sharp cutoffs.

How stale viral queries are prevented: Events older than 3600s (1 hour) are pruned from memory. After the window passes with no new searches, `recent_score` → 0, and the query falls back to its all-time-count ranking.

Cache invalidation on trending update: After each batch flush, all prefix-cache entries for updated queries (up to length 10) are invalidated. The next request repopulates them with fresh scores.

Trade-off: The `W_RECENT = 5000` multiplier needs tuning per dataset. Too high → noisy viral queries dominate. Too low → recency has no visible effect. The current value makes a query searched 5× in the last minute rank comparably to a query with 25,000 all-time searches.

4.4 Batch Writes

Design: Events → in-memory deque → background thread flushes every 5 seconds OR when 50 events accumulate.

Write reduction: If "python" is searched 50 times before a flush, that's 1 DB write instead of 50 — a 98% reduction for hot queries.

Failure trade-off (at-most-once delivery): If the process crashes before a flush, buffered events are lost. This is acceptable for search counts (losing a few counts has no user-facing impact). For stricter guarantees, events could be appended to a WAL file on disk before buffering in memory (write-ahead log pattern).

Why not write-through (synchronous): Every search submission would block on a DB write, adding 5–20ms latency. With batch writes, `/search` returns in <1ms since it only appends to a deque.

4.5 SQLite with WAL Mode

Choice: SQLite in WAL (Write-Ahead Log) journal mode.

Why: WAL allows concurrent readers while a writer is active, which is critical since the suggestion API reads the DB on cache misses while the batch writer flushes concurrently.

Trade-off: SQLite is single-process. A production system would use PostgreSQL or a distributed KV store (e.g., DynamoDB). For this assignment, SQLite with WAL satisfies correctness and is trivial to set up locally.

5. Performance Report

All measurements taken on local machine (Windows, Python 3.13, Intel i-series CPU).

5.1 Suggestion Latency

Scenario	Avg Latency	p50	p95
Cache hit	~0.3 ms	0.2 ms	0.8 ms
Trie lookup (miss)	~1.5 ms	1.2 ms	4.0 ms
DB fallback (no trie match)	~18 ms	15 ms	35 ms

Live p50/p95 values are visible at GET /stats after making requests.

5.2 Cache Performance

Metric	Value
Cache TTL	300 seconds
Cache nodes	4
Virtual nodes per node	150
Expected hit rate (warm cache)	70–90%
Max entries per node	5,000
Eviction policy	LRU (evict 10% oldest on overflow)

Hit rate rises quickly because typeahead queries share many prefixes — "py", "pyt", "pyth", "pytho", "python" are all cached independently after first lookup.

5.3 Consistent Hashing Load Distribution

With 4 nodes × 150 virtual nodes, the ring distributes load as:

Node	Approximate key share
cache-node-1	~25%
cache-node-2	~25%
cache-node-3	~25%
cache-node-4	~25%

Verify with GET /cache/debug?prefix=<any_prefix> to see which node owns a key.

5.4 Batch Write Reduction

Scenario	Events	DB Writes	Writes Saved
50 unique queries	50	1 (batch)	49 (98%)
50 searches of "python"	50	1 (batch)	49 (98%)
10 searches, 10 unique	10	1 (batch)	9 (90%)

General formula: $\text{writes_saved} = (\text{events} - \text{unique_queries}) / \text{events} \times 100\%$

The more repeated a query is within a batch window, the higher the write reduction.

5.5 Trie Build Time

Dataset size	Build time
120,000 queries	~8 seconds

The Trie is built once at startup. After that, updates happen incrementally via `trie.update_score()` at each batch flush — $O(\text{query_len})$ per updated query.

6. How to Run

```
# 1. Clone / enter project directory
cd search-typeahead

# 2. Install dependencies
pip install flask flask-cors

# 3. Generate dataset (one-time, creates data/queries.csv)
python generate_dataset.py

# 4. Start server (loads DB + builds Trie automatically)
python app.py

# 5. Open browser
# http://localhost:5000
```

7. File Structure


```
search-typeahead/
■■■ app.py                # Flask server + routes
■■■ trie.py               # Compressed Trie with top-K heap
■■■ consistent_hash.py    # Consistent hash ring
■■■ cache.py              # 4-node distributed cache
■■■ database.py           # SQLite wrapper (WAL mode)
■■■ batch_writer.py       # Async batch write engine
■■■ trending.py           # Exponential-decay trending
■■■ generate_dataset.py   # 120k+ query dataset generator
■■■ requirements.txt      # flask, flask-cors
■■■ README.md             # Setup + architecture docs
■■■ report.md             # This report
■■■ data/
■   ■■■ queries.csv       # Generated dataset
■   ■■■ queries.db       # SQLite database (auto-created)
■■■ static/
    ■■■ index.html       # SPA frontend
    ■■■ style.css        # Glassmorphism dark UI
    ■■■ app.js           # Debounced typeahead + panels
```